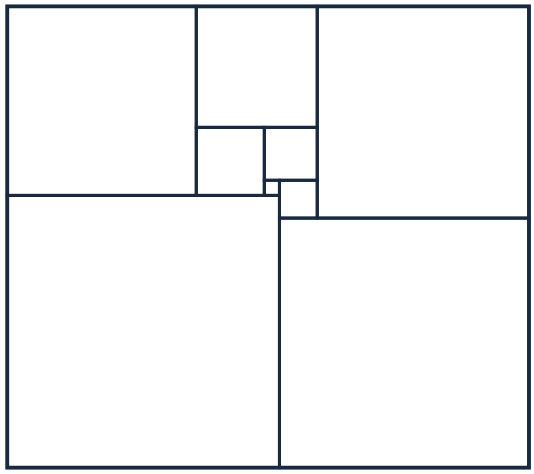
Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2000-I-Q04] | Part A (Q1-Q5)

A rectangle is tiled by nine squares in the arrangement shown. Its width and height are relatively prime positive integers. Find its perimeter.

一个矩形按图示方式分成九个正方形，宽和高为互质正整数。求矩形周长。



Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2004-II-Q05] | Part A (Q1-Q5)

A job is scheduled for 1000 equal-rate workers. After its first quarter, 100 leave; after the second quarter, another 100 leave. Once three quarters are finished, how many additional workers beyond the remaining 800 are minimally needed to meet the original deadline?

某工程原由一千名等效率工人按时完成。做完四分之一后离开一百人，再做完四分之一又离开一百人。完成四分之三时，除剩下的八百人外至少再招多少人，才能不超过原工期?

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2006-I-Q05] | Part A (Q1-Q5)

For positive integers a, b, c, $\sqrt{2006 + 104\sqrt{6} + 468\sqrt{10} + 144\sqrt{15}} = a\sqrt{2} + b\sqrt{3} + c\sqrt{5}$. Find abc.

正整数 a, b, c 满足 $\sqrt{2006 + 104\sqrt{6} + 468\sqrt{10} + 144\sqrt{15}} = a\sqrt{2} + b\sqrt{3} + c\sqrt{5}$，求 abc。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2007-I-Q02] | Part A (Q1-Q5)

A 100-foot walkway moves at 6 ft/s. Person A rides from time zero, B starts two seconds later walking at 4 ft/s relative to the walkway, and C starts two more seconds later beside it at 8 ft/s. After all start and before anyone finishes, one is midway between the others. Find that person’s distance from the start.

百英尺长的传送步道速度为六英尺每秒。甲在零时登上后站立，乙晚两秒登上并相对步道以四英尺每秒行走，丙再晚两秒沿旁边以八英尺每秒前行。三人都出发且尚未走完时，一人位于另两人正中，求此人的位置。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2007-II-Q04] | Part A (Q1-Q5)

Identical workers require fixed production times per item type. One hundred worker-hours produce 300 items of type X and 200 of Y; 120 worker-hours produce 240 X and 300 Y. With 150 worker-hours, producing 150 X leaves capacity for how many Y?

所有工人生产每种产品所需时间固定相同。一百工时可生产三百件甲及二百件乙；一百二十工时可生产二百四十件甲及三百件乙。一百五十工时生产一百五十件甲后，还能生产多少乙?

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2008-I-Q01] | Part A (Q1-Q5)

At a party, 60% are girls and 40% like dancing. Twenty additional boys, all dancers, arrive, reducing the girl fraction to 58%. How many attendees now like dancing?

聚会上女生占百分之六十，喜欢跳舞者占百分之四十。又来二十名男生且都喜欢跳舞，女生比例变为百分之五十八。求现有多少人喜欢跳舞。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2008-II-Q02] | Part A (Q1-Q5)

Cyclist R rests five minutes after every mile; cyclist J rides at three quarters R’s riding speed and rests five minutes after every two miles. They start together and first reach mile fifty simultaneously, before any final rest. Find elapsed minutes.

骑手甲每骑一英里休五分钟；乙骑行速度为甲的四分之三，每两英里休五分钟。两人同时出发且同时首次到达五十英里处，不计终点休息，求分钟数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2009-II-Q01] | Part A (Q1-Q5)

Initially there are $130, 164, 188$ ounces of blue,red,white paint. Four equal-area stripes use equal paint amounts: one of each pure color and one pink mixture of red and white. All three original colors have equal amounts left. Find total paint left.

原有蓝、红、白漆分别一百三十、一百六十四、一百八十八盎司，用等量油漆刷四条同面积条纹：蓝、红、白及红白混合粉色。之后三种原漆余量相等，求总余量。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Working / 解题过程

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2011-I-Q01] | Part A (Q1-Q5)

Jar A has 4 L at 45% acid and jar B has 5 L at 48%. Split 1 L of $k\%$ acid between them so both become 50% acid. If the amount added to A is reduced m/n liters, find $k+m+n$.

A 瓶有四升 45% 酸液，B 瓶有五升 48% 酸液。将一升 $k\%$ 酸液分入两瓶，使两瓶均为 50%。加入 A 的升数为最简分数 m/n，求 $k+m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2011-II-Q01] | Part A (Q1-Q5)

A drinker consumes the fraction m/n of a purchase. Had the purchase been half as large and consumption twice as large, the leftover would have been $2/9$ of the actual leftover. Find $m+n$ for the reduced fraction.

某人喝掉所购饮料的最简分数 m/n。若购买量减半、喝掉量加倍，剩余量只有实际剩余的 $2/9$。求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2012-I-Q04] | Part A (Q1-Q5)

Two travelers walk at 4 and 2.5 mph and share a horse traveling at 6 mph. The slower traveler initially rides, ties the horse at milepost one, and walks on. The other rides upon reaching the horse, passes the partner, and leaves the horse at the next milepost; they repeat this procedure. If their first meeting at a positive integer milepost is n miles from the start after t minutes, find $n + t$.

两人步行速度为每小时四英里、二点五英里，共用一匹每小时六英里的马。慢者先骑马，在一英里路桩拴马后步行；另一人到马处骑上，追过同伴后在下一路桩留下马，如此轮流。若首次在正整数英里路桩相遇时距起点 n 英里、历时 t 分钟，求 $n + t$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2012-II-Q04] | Part A (Q1-Q5)

Three cyclists start at a rectangle northeast corner: one follows the boundary westward at 8.6 m/s, one southward at 6.2 m/s, and one rides straight at 5 m/s. Their first arrivals coincide at a point D on the south edge. The west-east length exceeds the north-south width. Write length:width:distance from D to the southeast corner as $p : q : r$, with coprime p, q, and find $p + q + r$.

三名骑车人从矩形东北角同时出发，沿边向西者速率 8.6 米/秒，沿边向南者 6.2，直线骑行者为五。三人首次同时到达南边点 D。东西长大于南北宽；将长、宽、D 到东南角距离之比写为 $p : q : r$，其中 p, q 互质，求三者和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2013-I-Q01] | Part A (Q1-Q5)

An athlete swims $1/2$ mile, cycles thirty miles, and runs eight miles at constant speeds. Running is five times swimming speed, and cycling twice running speed. The total time is 4.25 hours. Find the cycling time in minutes.

运动员以各自恒定速度游半英里、骑车三十英里、跑八英里。跑速为游速五倍，骑速为跑速两倍，总时间四点二五小时。求骑车分钟数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2013-I-Q03] | Part A (Q1-Q5)

Two lines parallel to adjacent sides divide a square into two smaller squares and two nonsquare rectangles. The small squares together occupy $9/10$ of the original area. If the two segments of one cut side have lengths x, y, find $x/y + y/x$.

两条平行于相邻边的直线将正方形分成两个小正方形及两个非正方形矩形。小正方形总面积为原面积的十分之九。一条被分边的两段长为 x, y，求 $x/y + y/x$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2013-II-Q01] | Part A (Q1-Q5)

A decimal clock divides a day into ten hours of one hundred minutes each. Ordinary $6{:}36$ AM becomes $A{:}BC$. Find $100A + 10B + C$.

十进制时钟将一天分成十小时，每小时一百分。通常的早晨六点三十六分显示为 $A{:}BC$，求 $100A + 10B + C$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2014-I-Q04] | Part A (Q1-Q5)

Two cyclists ride east and west at twenty mph. Two equal-length trains travel oppositely at different constant speeds, overtaking a cyclist when moving in the same direction. Each takes one minute to pass the eastbound cyclist; the westbound train takes ten times as long as the eastbound train to pass the westbound cyclist. Write the train length as reduced m/n miles and find $m+n$.

两名骑车人分别向东、向西，速度均为每小时二十英里。两列等长列车方向相反、速率不同，并会超越与其同向的骑车人。每列通过东行者均需一分钟；西行列车通过西行者的时间是东行列车的十倍。若列车长度为最简分数 m/n 英里，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2014-II-Q01] | Part A (Q1-Q5)

Worker A paints a room in fifteen hours; B works fifty percent faster and C twice as fast as A. First A works ninety minutes alone, then A, B work until half the room is done, then all three finish. Find the total minutes.

甲独刷一房需十五小时，乙效率比甲高百分之五十，丙为甲两倍。甲先单独工作九十分钟，再与乙一起刷到一半，最后三人共同完成。求总分钟数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2017-II-Q05] | Part A (Q1-Q5)

Four distinct real numbers have six unordered pair sums, listed without order as $189, 320, 287, 234, x, y$. Find the largest possible $x + y$.

四个不同实数的六个无序两两和，未按顺序列为 $189, 320, 287, 234, x, y$。求 $x + y$ 的最大值。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2018-II-Q01] | Part A (Q1-Q5)

Points A, B, C lie in order on a straight path with $AC = 1800$ m. Runners start together: Ina at A toward C, Paul at B toward C, Eve at C toward A, with speed ratio Ina:Paul:Eve $= 2 : 4 : 1$. Paul turns back when he meets Eve and returns to B exactly as Ina first reaches it. Find AB.

直路上 A, B, C 依次排列，$AC = 1800$ 米。三人同时出发，伊娜从 A 向 C、保罗从 B 向 C、伊芙从 C 向 A，速度比为 $2 : 4 : 1$。保罗遇到伊芙后折返，恰与伊娜首次同时到达 B。求 AB。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2022-I-Q05] | Part A (Q1-Q5)

A river of width 264 m flows east at 14 m/min. Two swimmers start on the south bank, the faster one D m east of the slower. Their speeds relative to water are 80 and 60 m/min. Both swim straight to the same north-bank point, equidistant from their starts, and arrive together. Find D.

河宽 264 米，水流向东，速度 14 米/分。两人从南岸出发，较快者在较慢者以东 D 米；静水速度分别为 80、60 米/分。两人沿直线游向北岸同一点，该点到两出发点等距，且同时到达。求 D。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2023-II-Q04] | Part A (Q1-Q5)

Real numbers satisfy $xy + 4z = yz + 4x = zx + 4y = 60$. Let S contain every possible value of x. Compute $\sum_{s \in S} s^2$.

实数 x, y, z 满足 $xy + 4z = yz + 4x = zx + 4y = 60$。令 S 为 x 的所有可能值组成的集合，求各元素平方和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2024-I-Q01] | Part A (Q1-Q5)

A daily outing consists of a 9 km walk and a fixed stop of t minutes. At walking speed s km/h the total is 4 h; at $s+2$ km/h it is 144 min. Find the total minutes at speed $s+1/2$ km/h.

一次日常外出包括步行 9 千米及固定停留 t 分钟。步速为 s 千米/时时总用时 4 小时；步速为 $s+2$ 时总用时 144 分钟。求步速为 $s+1/2$ 时的总分钟数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2026-I-Q01] | Part A (Q1-Q5)

Three travelers follow the same route and arrive together. The second starts one hour after the first at 2 mph faster; the third starts one hour after the second at 7 mph faster than the second. Write the route length as a reduced fraction m/n and find $m+n$.

三人走同一路线并同时到达。第二人比第一人晚一小时出发，速度快 2 英里/时；第三人又晚一小时出发，速度比第二人快 7 英里/时。路程为最简分数 m/n 英里，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2000-I-Q07] | Part B (Q6-Q10)

Positive real numbers satisfy $xyz = 1$, $x + 1/z = 5$, and $y + 1/x = 29$. Write $z + 1/y = m/n$ reduced and find $m + n$.

正实数满足 $xyz = 1, x + 1/z = 5, y + 1/x = 29$。将 $z + 1/y$ 约分为 m/n，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2000-I-Q10] | Part B (Q6-Q10)

For $k = 1, \ldots, 100$, a list $x_1, \ldots, x_{100}$ satisfies $x_k = \sum_{j \neq k} x_j - k$. If $x_{50} = m/n$ reduced with $m, n > 0$, find $m + n$.

一列数满足：对 $k = 1$ 至 100，$x_k = \sum_{j \neq k} x_j - k$。若 x_{50} 约分为正整数之比 m/n，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2010-I-Q09] | Part B (Q6-Q10)

Real x, y, z satisfy $x^3 - xyz = 2, y^3 - xyz = 6, z^3 - xyz = 20$. The maximum of $x^3 + y^3 + z^3$ is the reduced fraction m/n. Find $m + n$.

实数满足 $x^3 - xyz = 2, y^3 - xyz = 6, z^3 - xyz = 20$。$x^3 + y^3 + z^3$ 的最大值为最简分数 m/n，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2011-I-Q06] | Part B (Q6-Q10)

A parabola $y = ax^2 + bx + c$ has vertex $(1/4, -9/8)$, $a > 0$, and integer $a + b + c$. If the least possible a is reduced p/q, find $p + q$.

抛物线 $y = ax^2 + bx + c$ 的顶点为 $(1/4, -9/8)$，$a > 0$，且 $a + b + c$ 为整数。若 a 的最小值为最简分数 p/q，求 $p + q$。

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Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2021-II-Q07] | Part B (Q6-Q10)

Real numbers a, b, c, d satisfy $a+b=-3$, $ab+bc+ca=-4$, $abc+bcd+cda+dab=14$, and $abcd=30$. If $a^2+b^2+c^2+d^2=m/n$ reduced with $m, n>0$, find $m+n$.

实数 a, b, c, d 满足 $a+b=-3$、$ab+bc+ca=-4$、$abc+bcd+cda+dab=14$、$abcd=30$。若平方和约分为 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2001-I-Q11] | Part C (Q11-Q15)

A $5 \times N$ point array is numbered first by rows, then renumbered by columns. Choose P_i in row i, with old number x_i and new number y_i. Require $x_1 = y_2, x_2 = y_1, x_3 = y_4, x_4 = y_5, x_5 = y_3$. Find the least N.

一个五行 N 列点阵先逐行编号，再逐列重新编号。在第 i 行选点 P_i，原编号为 x_i、新编号为 y_i。要求 $x_1 = y_2, x_2 = y_1, x_3 = y_4, x_4 = y_5, x_5 = y_3$，求最小 N。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2014-I-Q14] | Part C (Q11-Q15)

Let m be the greatest real solution of $3/(x-3)+5/(x-5)+17/(x-17)+19/(x-19)=x^2-11x-4$. If $m=a+\sqrt{b+\sqrt{c}}$ for positive integers, find $a+b+c$.

令 m 为方程 $3/(x-3)+5/(x-5)+17/(x-17)+19/(x-19)=x^2-11x-4$ 的最大实根。若 $m=a+\sqrt{b+\sqrt{c}}$，求正整数 $a+b+c$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2015-II-Q14] | Part C (Q11-Q15)

Real numbers x, y satisfy $x^4y^5 + y^4x^5 = 810$ and $x^3y^6 + y^3x^6 = 945$. Evaluate $2x^3 + (xy)^3 + 2y^3$.

实数 x, y 满足 $x^4y^5 + y^4x^5 = 810$、$x^3y^6 + y^3x^6 = 945$。求 $2x^3 + (xy)^3 + 2y^3$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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